Mathematical Analysis Homework Week 11

Tan Renhao

2024.5

1. 举例说明: E 是 $\mathbb{R}^n$ 中的有界开集, 但 ∂E 不是零测集, 则 χ_E 在 E 上广义可积, 但在通常的 Riemann 积分意义下是不可积的.

   证明. 记 $[0,1]$ 之间的所有有理数为 $\{q_1, q_2, \ldots\}$. 取 $\varepsilon > 0$, 令

   $$E := \bigcup_{k=1}^{\infty} \left(q_k - \frac{\varepsilon}{2^{k+1}}, q_k + \frac{\varepsilon}{2^{k+1}}\right)$$

   则 E 为有界开集, 从而 χ_E 在其上广义可积.

   任意 $\forall x \in [0,1]\backslash E$ 可以被有理数逼近, 从而 $[0,1]\backslash E \subseteq \partial E$. 故 $m(\partial E) \geq 1 - \sum\limits_k \varepsilon/2^k = 1 - \varepsilon$, 于是 E Jordan 不可测, 也即 χ_E Riemann 不可积. □

2. 计算积分

   (a)

   $$\begin{aligned}\iint_{y>x^2+1} \frac{\mathrm{d}x\,\mathrm{d}y}{x^4+y^2} =& \int_{-\infty}^{+\infty} \mathrm{d}x \int_{x^2+1}^{+\infty} \frac{1}{x^4+y^2}\,\mathrm{d}y = \int_{-\infty}^{+\infty} \mathrm{d}x \left[\frac{1}{x^2}\arctan\frac{y}{x^2}\right]_{x^2+1}^{+\infty} \\ =& 2\int_0^{+\infty} \frac{1}{x^2}\left(\frac{\pi}{2} - \arctan\left(1+\frac{1}{x^2}\right)\right)\mathrm{d}x \\ =& 2\int_0^{+\infty} \frac{\pi}{2} - \arctan\left(1+t^2\right)\mathrm{d}t \\ =& 2\left[\left(\frac{\pi}{2} - \arctan\left(1+t^2\right)\right)t\right]_0^{+\infty} + 4\int_0^{+\infty} \frac{t^2}{\left(1+t^2\right)^2+1}\,\mathrm{d}t\end{aligned}$$

   根据 L'Hospital 法则

   $$\lim_{t\to+\infty}\left(\frac{\pi}{2} - \arctan\left(1+t^2\right)\right)t = \lim_{t\to+\infty} \frac{-\frac{2t}{(1+t^2)^2+1}}{-1/t^2} = \lim_{t\to+\infty} \frac{2t^3}{\left(1+t^2\right)^2+1} = 0,$$

   而

   $$\begin{aligned}\int_0^{+\infty} \frac{t^2}{\left(1+t^2\right)^2+1}\,\mathrm{d}t =& \int_0^{+\infty} \frac{t^2}{(t^2+\sqrt{2}-\sqrt{2\sqrt{2}-2}t)(t^2+\sqrt{2}+\sqrt{2\sqrt{2}-2}t)}\,\mathrm{d}t \\ =& \frac{1}{2\sqrt{2\sqrt{2}-2}}\int_0^{+\infty} \frac{t}{t^2+\sqrt{2}-\sqrt{2\sqrt{2}-2}t} - \frac{t}{t^2+\sqrt{2}+\sqrt{2\sqrt{2}-2}t}\,\mathrm{d}t\end{aligned}$$

$$=\frac{\pi}{4}\sqrt{2\sqrt{2}-2}$$

因此

$$\iint_{y>x^2+1}\frac{\mathrm{d}x\ \mathrm{d}y}{x^4+y^2}=2\left[\left(\frac{\pi}{2}-\arctan\left(1+t^2\right)\right)t\right]_0^{+\infty}+4\int_0^{+\infty}\frac{t^2}{\left(1+t^2\right)^2+1}\ \mathrm{d}t=\pi\sqrt{2\sqrt{2}-2}$$

(b)

$$\begin{aligned}\iint_{x^2+y^2<1}\log\frac{1}{\sqrt{x^2+y^2}}\ \mathrm{d}x\ \mathrm{d}y&=\int_0^1\int_0^{2\pi}r\log\frac{1}{r}\ \mathrm{d}r\ \mathrm{d}\theta=2\pi\int_0^1-r\log r dr\\&=\pi\left(-r^2\log r\Big|_0^1+\int_0^1 r dr\right)=\frac{\pi}{2}\end{aligned}$$

(c)

$$\begin{aligned}\iint_{x^2+y^2<1}\frac{\mathrm{d}x\ \mathrm{d}y}{\left(1-x^2-y^2\right)^\alpha}&=\int_0^1\int_0^{2\pi}\frac{r}{\left(1-r^2\right)^\alpha}drd\theta=2\pi\int_0^1\frac{r}{\left(1-r^2\right)^\alpha}dr\\&=\pi\int_0^1\frac{1}{x^\alpha}dx=\frac{\pi}{1-\alpha}\quad(\alpha<1)\end{aligned}$$

(d) 作换元

$$(x,y)=\varphi(r,\theta)=\left((r\cos\theta)^{2/p},(r\sin\theta)^{2/q}\right)$$

那么

$$\det\mathbf{J}\varphi=\begin{vmatrix}\frac{2}{p}\frac{(r\cos\theta)^{2/p}}{r} & -\frac{2}{p}(r\cos\theta)^{2/p}\tan\theta\\ \frac{2}{q}\frac{(r\sin\theta)^{2/q}}{r} & \frac{2}{q}\frac{(r\sin\theta)^{2/q}}{\tan\theta}\end{vmatrix}=\frac{2}{pq}\frac{(r\cos\theta)^{2/p}(r\sin\theta)^{2/q}}{r\cos\theta\sin\theta}$$

代入得

$$\begin{aligned}\iint_{|x|^p+|y|^q>1}\frac{\mathrm{d}x\ \mathrm{d}y}{|x|^p+|y|^q}&=4\iint_{x^p+y^q>1,x>0,y>0}\frac{\mathrm{d}x\ \mathrm{d}y}{x^p+y^q}\\&=4\int_1^\infty\int_0^{2\pi}\frac{2}{pq}\frac{(r\cos\theta)^{2/p}(r\sin\theta)^{2/q}}{r^3\cos\theta\sin\theta}drd\theta\\&=\frac{8}{pq}\int_0^{\pi/2}\cos^{2/p-1}\theta\sin^{2/q-1}\theta\mathrm{d}\theta\int_1^{+\infty}r^{2/p+2/q-3}\ \mathrm{d}r\\&=\frac{4}{pq}\frac{1}{1-\left(\frac{1}{p}+\frac{1}{q}\right)}B\left(\frac{1}{p},\frac{1}{q}\right)\end{aligned}$$

3. 设 E_1,E_2 为 $\mathbb{R}^n$ 中的开集, $f:E_1\cup E_2\to\mathbb{R}$ 连续, 且 f 在 E_1,E_2 上绝对可积, 则 f 在 $E_1\cup E_2$ 以及 $E_1\cap E_2$ 上也绝对可积, 且

$$\int_{E_1\cup E_2}f(x)\mathrm{d}x=\int_{E_1}f(x)\mathrm{d}x+\int_{E_2}f(x)\mathrm{d}x-\int_{E_1\cap E_2}f(x)\mathrm{d}x.$$

证明. 取两列 Jordan 可测的紧集 $\{A_i\},\{B_i\}$ 分别逼近 E_1,E_2. 则

$$\{A_i\cap B_i\}\to E_1\cap E_2,\quad\{A_i\cup B_i\}\to E_1\cup E_2$$

由于 f 在 E_1,E_2 上均绝对可积

$$\int_{A_i\cup B_i}|f(x)|\mathrm{d}x+\int_{A_i\cap B_i}|f(x)|\mathrm{d}x=\int_{A_i}|f(x)|\mathrm{d}x+\int_{B_i}|f(x)|\mathrm{d}x.$$

从而

$$\int_{A_i\cup B_i} f(x)\mathrm{d}x + \int_{A_i\cap B_i} f(x)\mathrm{d}x = \int_{A_i} f(x)\mathrm{d}x + \int_{B_i} f(x)\mathrm{d}x,$$

令 $i \to \infty$, 我们有

$$\int_{E_1\cup E_2} f(x)\mathrm{d}x + \int_{E_1\cap E_2} f(x)\mathrm{d}x = \int_{E_1} f(x)\mathrm{d}x + \int_{E_2} f(x)\mathrm{d}x.$$

□